(L # 17 starts)

Derivation of fill-rate eqn.

From eqn. (3) for the fill-rate section:

$$f_r = \frac{\text{expected demand}}{\text{actual demand}} = \frac{Q_L}{Q_L + ESC} \rightarrow (3) \text{ repeated}$$

(during the replenishment cycle time, T)

$$f_r = \frac{1}{1 + \left(\frac{ESC}{Q_L}\right)} \rightarrow (4)$$

$$x \triangleq \frac{ESC}{Q_L} \rightarrow (5)$$

$$f_r \underset{(4,5)}{=} \frac{1}{1+x} \rightarrow (6)$$

If $x \ll 1$, $x^2 \rightarrow 0$

(example, $x = 0.01$, then $x^2 = 0.0001 \approx 0$)

$\therefore$ for $x \ll 1$, $(1+x)(1-x) = 1 - x^2 \approx 1$

$$\Rightarrow 1 + x = \frac{1}{1-x}, \text{ or } \frac{1}{1+x} = 1 - x \rightarrow (7)$$

$$\Rightarrow f_r \underset{6,7}{=} 1 - x \underset{(5)}{=} 1 - \left(\frac{ESC}{Q_L}\right)$$

$$\boxed{\text{Fill rate} = 1 - \left(\frac{ESC}{Q_L}\right) = \frac{Q_L - ESC}{Q_L}} \rightarrow (8)$$

TIM 172 B, MOT-II (SCM), 3/3/2026
Lecture # 17

Agenda:

1. Supplier Lead-time uncertainty & Safety Inventory
2. Total Inventory Holding cost
3. Inventory Management Summary
4. Numerical calculations
5. Facilities
6. Transportation → posted a set of lecture notes on Canvas

1. Supplier Lead-Time Uncertainty

(HW # 7, Prob. 3)

Notation :

(i) Uncertainties in weekly demand : $\underbrace{D_W, \sigma_W}_{\text{as before}}$

(ii) Uncertainties in demand during the supplier lead-time, L

$$\underbrace{(D_L)_m = L\, D_W \; ; \; \sigma_L = \sqrt{L}\, \sigma_W}_{\text{as before}}$$

(iii) supplier lead-time is characterized by a mean, L (weeks), and a standard deviation $= s_d$ (weeks)

Convert the std. dev., s_d (weeks) to an equivalent demand,

If D_W is the expected weekly demand, then the demand during s_d weeks $= (s_d)(D_W) \rightarrow (1)$

$$[\text{weeks}]\left[\frac{\text{units}}{\text{week}}\right]$$

Recall that <u>variances are additive</u>

To take uncertainties in both demand & uncertainties in supplier lead time into account, compute

$$\sigma_{Agg}^2 \underset{(1)}{=} \sigma_L^2 + (s_d D_w)^2$$

$$\sigma_{Agg} = \sqrt{(\sigma_L)^2 + (s_d D_w)^2} \longrightarrow (2)$$

<u>Injunction</u> : To solve all safety inventory problems for which both customer demand & supplier lead-time are uncertain, replace σ_L in all <u>previous</u> continuous review eqns. with σ_{Agg}, given by eqn. (2) above.

For example,

(1) $CSL = F\left(z' = \frac{ss}{\sigma_L}\right)$ for continuous review

becomes :

$CSL = F\left(z' = \frac{ss}{\sigma_{Agg}}\right)$ for continuous review with uncertainties in both customer demand & supplier lead-time.

(ii) $$\frac{ss}{\sigma_L} = F^{-}\left[(CSL)_{desired}\right]$$

becomes

$$\frac{ss}{\sigma_{Agg}} = F^{-1}\left[(CSL)_{desired}\right],$$

ETC.

For HW # 7, Prob. 3

Given $(CSL)_{desired} = 95\%$ or 0.95

& other data

Determine, ss, ...

Result : for $s_d = 1.5$ weeks, $ss \approx 709$ items

Total Inventory Holding Cost (Annual)

$$\left.\begin{array}{l}\text{Annual cycle inventory}\\ \text{holding cost}\end{array}\right\} = \left(\begin{array}{c}\text{average or}\\ \text{cycle inv.}\end{array}\right)\left(\begin{array}{c}\text{holding cost}\\ \text{per unit per YR.}\end{array}\right)$$

$$= \left(\frac{Q_L}{2}\right)(hC) \rightarrow (1)$$

$$\left.\begin{array}{l}\text{Annual safety inventory}\\ \text{holding cost}\end{array}\right\} = (\text{safety stock})\left(\begin{array}{c}\text{holding cost}\\ \text{per unit}\\ \text{per YR.}\end{array}\right)$$

$$= (SS)(hC) \rightarrow (2)$$

$$\left.\begin{array}{l}\text{Total inventory}\\ \text{holding cost}\end{array}\right\} \underset{(1,2)}{=} \left(\frac{Q_L}{2} + SS\right)(hC) \rightarrow (3)$$

These costs need to be calculated for all relevant inventory for your company's product for all years of interest.

⇒ Automate with EXCEL
⇒ Present compactly in tables

3. Summary of Inv. Mgmt.

Cycle Inventory $= \frac{Q_L}{2}$,
(average)

is all about efficiency,
as in economies of scale:

Determine the optimal
lot size, Q_L^*, to
minimize total annual cost

Time Horizon: 1 year
$\Rightarrow$ use annual demand, D,
to optimize lot size.

$\Rightarrow$ PLANNING LEVEL

Safety Inventory, ss
is all about
responsiveness, as in

Is the product available
to all potential
customers when needed:

Determine the safety
stock, ss, to meet
some desired value
of a product availability
metric:
$(CSL)_{desired}$; $(f_r)_{desired}$

Time Horizon:
supplier lead-time:
L weeks
weekly demand statistics:
D_W, σ_W

$\Rightarrow$ OPERATIONAL LEVEL

4. Doing numerical calculations

Here's what we see in your numerical work

$D(\text{year } 3, \text{ qtr. } 1) = \underbrace{1379476.12}_{\text{too many significant digits}}$

Counting significant digits :

The first significant digit is the first non-zero digit in the number :

Example : 0.00032205

non-significant digits | first significant digit | 2nd sig digit

Representing numbers

0.0003205 ⟶ 0.3205 (10^{-3})
.3205 E-03

exponential or scientific notation

Guidelines :

(i) Always write numbers in exponential scientific notation

(ii) Then count the number of significant digits

(iii) Use 3, at most 4, significant digits

Suggestion :

Use 4 significant digits for calculations, & 3 significant digits for the result

$\therefore \underbrace{897342.77}_{\text{too many significant digits}} \longrightarrow$ 897300. (for calculations)

897000. (for results

.897 E06 } scientific notation

$.897(10^6)$

Next two important topics

Facilities → Lecture #18 (Thursday)

Transportation → refer to the Transportation lecture notes on Canvas

Optimization at a Large Scale

MS Solver : Study the "MS Excel Solver: Quick start" handout on Canvas:

Injunction : do the Tutorial in the handout